Vitalii Ryzhenko

Applied AI / Forward Deployed Engineer | Full-Stack TypeScript · .NET · Python | LLM Features, MCP, Agent Evals

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Summary

Applied AI engineer with 9+ years of full-stack experience across enterprise, healthcare and e-commerce platforms (TypeScript, Angular/React, .NET/Node). I ship LLM features and AI-agent tooling to production — MCP tools, structured outputs, context engineering and agent evals — and work directly with customers, taking vague problems to production fixes. Before engineering I worked in B2B sales — closing large deals and leading negotiations — so discussing scope, cost and value with customers comes naturally.

Technical Skills

- **AI & LLM:** Claude Code (skills, hooks, sub-agents, CLAUDE.md, MCP servers), Model Context Protocol (tool design, contracts, E2E testing), LLM coding agents (skills, hooks, sub-agents, parallel sessions), Structured output via JSON Schema, Prompt & context engineering, Agent eval scenarios and A/B runs, OpenAI API integration and an analytics agent in production at Creatio
- **Frontend:** TypeScript, Angular — 8+ years (DI, router, lazy modules, change detection), RxJS, NgRx, React, Redux, Angular Material/CDK, AG Grid, Chart.js, Web Components, Storybook, SCSS, Bootstrap
- **Backend & Data:** C#/.NET, Node.js, Python, REST APIs, MS SQL, PostgreSQL
- **Build & Delivery:** Nx monorepo, Webpack/Module Federation, Jenkins, GitLab CI, Docker, AWS, Azure, SonarQube, Feature flags, Release-line backporting
- **Testing:** Jest + TestBed, Playwright (page objects, test-framework SDK), NUnit, Cypress, sitespeed.io
- **Practices:** WCAG/ARIA accessibility, i18n, L4 production support, ADRs and technical documentation

Experience

Creatio — Senior Full-Stack Engineer

09.2023 – 10.2026 · B2B · Large Angular/TypeScript low-code platform (Nx monorepo, ~20 repositories, C#/.NET)

AI & agent tooling

- Built MCP tools for an open-source agent integration with the platform, incl. a deterministic nested-filter builder with schema validation that rejects invalid configurations before save; 23 merged PRs to the public clio, clio-knowledge and creatio-ai-app-development-toolkit repositories (as of Nov 2026).
- Replaced the agent's overloaded always-on instructions with a thin router and on-demand guidance via tool descriptions and versioned bundles — cut persistent context ~75% and fit it under the client's truncation limit.
- Extended the agent eval suite with end-to-end scenarios that verify the produced result in a real browser (Playwright), with tool whitelists and negative cases.
- Delivered LLM-driven generation of analytics widgets in chat (5 widget types): JSON-Schema structured output, response-to-config converters, prompt engineering.
- Hardened LLM response handling in the product AI chat with deterministic post-processing and explicit errors — eliminated one defect class (backported to a supported release) and diagnosed another for the owning team.
- Embedded web-client mode running the AI assistant inside an Outlook add-in: built its routing and standalone library (own CI, tests), prototyped cookie-less authentication, co-built fault-tolerant session-token refresh. Add-ins for two host apps run on this mode.
- Built an agent-skill orchestrator with ~15 contract tests and a personal agentic workflow (pre-commit root-cause hooks, parallel agent sessions, reusable review/refactoring skills).

Customer escalations (L4)

- Handled ~175 L4 client escalations (2024–2026): reproduced on clean environments, traced to root cause, delivered a fix, workaround or justified closure; median time to close ~1 week.
- Untangled multi-layer defects (e.g. two bugs in designer and widget masking each other; a four-step metadata failure chain that froze home pages) and backported fixes to supported release lines.
- ~40% closed without product changes: separated product defects from configuration, environment and cross-team issues, with evidence for the owning team or clear guidance for the customer.
- Coordinated delivery across several teams: prepared task lists for dependent teams and drove the cross-team communication.

Front-end performance & architecture

- Led a team refactor splitting the monolithic analytics-widget library into a shared core and 6–7 lazy-loaded modules, removing chart code from the initial bundle (~0.5 MB, ~5% of always-loaded chunks).
- Moved a heavy code editor out of the initial bundle into a lazy-loaded library with its own CI, Storybook and tests.
- Parallelized web-client startup and added page-load telemetry; added a feature-flagged module-loading cache; backported background-tab load acceleration to two release lines.
- Migrated ~190 UI-library components to a new reuse strategy in ~2 weeks; removed redundant change-detection work and duplicate HTTP calls.

Product features

- Built analytics widgets and settings (metrics, calculated metrics, pivot tables); reduced database load from analytics queries; added pivot-table support and CLI delivery to the dashboard migrator.
- Owned technical research for WhatsApp Business template management via Twilio (two UI prototypes, architecture decision record); implemented the template lifecycle in a decoupled module with ~50 unit tests.
- Implemented WCAG accessibility (screen-reader announcements, focus management, keyboard indicators) and localization into dozens of languages.
- Published the developer SDK npm package with automated dependency-integrity checks; remediated dependency and container-image vulnerabilities.

Quality

- Raised share of commits with tests from 27% to 53% and added ~1K test cases; wrote Playwright E2E specs and ~13 test-framework SDK modules; reviewed ~185 pull requests in 2026.

DataArt — Senior Software Engineer

2019 – 2023

- A US benefits-tech startup (insurance benefits management): built the product's web application with React and Redux.
- E-commerce platform: front-end development with Angular, NgRx and Bootstrap; E2E tests with Cypress; CI/CD with Jenkins and Azure.
- A US telehealth provider: built a proof of concept on AWS and supported a webOS IoT device connecting patients and doctors — real-time health metrics, live patient communication and multi-doctor consultations.

Ciklum — Middle JavaScript Developer

2017 – 2019

- Automotive fleet CRM system: front-end development with Angular, NgRx and an Nx monorepo on a .NET backend; data-heavy UI with AG Grid and 3D visualization with Three.js.
- Front-end performance and test automation (Cypress, Jest).

Earlier — Frontend Developer, 2016–2017 (Lanet Network, BuddhaSoft; Angular 2+, HTML/CSS)

Education

Master's Degree, Management of Organizations and Administration — Volodymyr Dahl East Ukrainian National University · Professional Software Engineering Retraining (2017)

Certifications

DeepLearning.AI: Spec-Driven Development with Coding Agents · Anthropic Academy: Claude Platform 101, Introduction to Model Context Protocol, AI Fluency: Framework and Foundations, Claude Code 101, Introduction to subagents, Introduction to agent skills, Claude Code in Action, Claude 101

Languages

English (B2) · Polish (conversational)